

Takeaways

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Abstract

Predicting wildfires is next to impossible.

Keywords forest health, climate, predict utilization

- **Fire-adapted forests** are experiencing widespread stress, mortality, and megafires as a result of a century of fire suppression. Suppression has led to excessive fuel buildup and unnaturally dense forests, with too many trees competing for limited water and nutrients.

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1. BACKGROUND

1.a. Burning & air quality

We acknowledge that older biomass plants were dirty, but modern regulations, controls, and siting are helping to mitigate that issue. These and environmental justice impacts from siting facilities in disadvantaged communities can and should be further strengthened with strong CEQA and other environmental policies. Nevertheless, poor air quality and greenhouse gas emissions from wildfires dwarf regulated point sources. The nearly million-acre Dixie fire, California's second-largest wildfire, emitted 37 million tonnes of CO_{2e} or the equivalent emissions from electricity consumption of nearly 5 million households (Baldassare, 2024). In 2020, wildfires emitted 127 million tons of CO_{2e}, nearly twice California's total greenhouse gas emissions reductions achieved since 2003 (Jerrett et al., 2022).

Wolf suggests leaving biomass residues in the forest. Masticating on site, lopping and piling, or removing residue to landings has long been practiced at thinning sites. But the accumulation of materials is overwhelming. A study showed that there are approximately 150,000 acres of piles representing 1,000,000 tons, with 78% of these materials on public lands (Darlington et al., 2023).¹ Even disposing of those piles by burning is an episodic, unregulated process that would be much more harmful than utilization in long-term wood products or burning at a facility.

idize supply, biomass is a waste product. When we subsidize demand, biomass is a value proposition. (Clarke Stevenson, personal communication, 2026)

Large-scale forest treatments inevitably generate substantial volumes of woody material. From an economic standpoint, this material is not optional; it is an unavoidable byproduct of necessary restoration. If it is left in place, piled, or

¹Only 19% of this material is accessible via existing road networks.

masticated, wildfire risk is often shifted rather than reduced, and carbon is released through decay or open burning without benefit. That biomass must go somewhere. Deliberate utilization converts an unavoidable cost into a managed material flow that supports risk reduction rather than undermining it.

Another longer-term economic value is often undercounted: carbon storage and avoided emissions. Using forest biomass in longer-lived products such as mass timber, engineered wood, or biochar effectively sequesters carbon that would otherwise be released through wildfire or decomposition, while simultaneously displacing more emissions-intensive materials. Modeling studies show that treated forest scenarios can result in greater total carbon storage over time despite substantially lower tree densities, due to reduced wildfire emissions and increased growth of large trees (Delyser et al., 2025; Elias et al., 2025).

2. WILDFIRE DISINFORMATION

As with climate change, misinformation about wildfire, forests, and logging has increasingly shaped public understanding and policy debates (Jones et al., 2022). In California, this misinformation is often reinforced by deeply ingrained visual and cultural assumptions about what a healthy forest looks like. Many people implicitly picture dense, closed-canopy forests modeled on the eastern United States, northern Europe, or England, landscapes that evolved under very different climatic and ecological conditions.

California's forests exist largely within a Mediterranean climate, characterized by wet winters and long, hot, dry summers. These conditions favor fire-adapted ecosystems rather than the dense, moisture-rich forests common in temperate eastern regions. When forest health is judged using the wrong ecological reference point, fire-adapted landscapes can be misdiagnosed as degraded or overmanaged, while overly dense stands are mistakenly viewed as natural or desirable.

These misconceptions shape how wildfire risk is interpreted and how solutions are evaluated. Simplified narratives, such as the claim that destructive fires are primarily a shrubland phenomenon, further obscure the reality that California's most damaging fires occur across mixed landscapes that include forests, shrublands, and the wildland–urban interface. When wildfire ecology is misunderstood in this way, forest management and biomass utilization are often framed as unnecessary or harmful rather than as responses to a misaligned, historically altered system.

Wildfire disinformation matters not only because it is inaccurate and undermines accepted science, but because it drives policy choices that fail to address the underlying causes of high-severity fire in a fire-adapted landscape (Cook, 2020).

2.a. Fire-adapted forests & cultural fire

A growing body of ecological research, supported by historical records and Indigenous knowledge, makes clear that most of California's landscapes burned regularly prior to Euro-American settlement. From the coast to the Sierra Nevada and beyond, fire occurred on highly variable but frequent cycles, often on the order of five

to fifteen years, depending on vegetation type, elevation, and local climate. Fire was not an anomaly in these systems; it was essential to their function.

Indigenous peoples actively shaped these fire regimes through intentional burning, now commonly referred to as cultural burning. These practices did not simply mimic lightning ignitions but actively guided fire to support food systems, medicinal plants, wildlife habitat, travel corridors, and sacred landscapes. Cultural burning worked in concert with natural fire to sustain open, patchy forests, interspersed meadows, and diverse age structures, supporting high levels of biodiversity and resilience.

Decades of fire science align closely with this knowledge. Prior to widespread fire suppression, western forests tended to have larger trees, lower overall density, and greater structural heterogeneity (North et al., 2009; Stephens et al., 2007). The removal of frequent, low- to moderate-severity fire disrupted these systems, allowing fuels to accumulate and forest structure to shift far beyond historical and ecological norms.

2.b. Fire suppression

The transition from frequent, smaller fires to the large, high-severity wildfires of today is not a natural evolution but the result of more than a century of fire suppression (Collins et al., 2017; Peery et al., 2019). Historically, regular fire functioned as an active form of landscape management. It improved tree health by reducing competition, clearing understory fuels, removing dead or weakened vegetation, and opening the forest canopy to sunlight. These processes reduced water stress, improved nutrient cycling, and supported the development of meadows and open patches that acted as natural fire breaks.

Frequent fire also interacts with hydrology. By reducing excessive tree density, fire helped balance water demand with available moisture, supporting smaller, localized water cycles and maintaining soil and vegetation health. In these landscapes, most fires burned at lower intensity and often self-extinguished as they encountered patchy fuels and open areas. Large, stand-replacing fires were relatively rare.

Fire suppression reversed these dynamics. Forests became denser and more uniform, fuels accumulated continuously across large areas, and the patchiness that once limited fire spread was lost. Under modern climate conditions, this produced fires that burn hotter, spread farther, and cause far greater ecological and social damage than the frequent fires they replaced.

Restoring fire-adapted landscapes, therefore, requires reintroducing fire as a management tool, alongside thinning and other treatments that help re-establish appropriate structure. In the near term, this process generates large volumes of forest residuals. Biomass utilization does not drive this work; rather, it emerges from it as a practical necessity. Over time, as landscapes regain balance and fire resumes its ecological role, the need for intensive intervention and large-scale biomass handling can and should decline.

3. SOLUTIONS

Returning to the mantra burn, bury, build, we recommend that all biomass utilization solutions incorporate the highest and best uses of forest product pathways. Moving forward, several integrated pathways can support forest health and biomass utilization:

1. **Climate adaptation and biomass integration.** Rapid scaling of forest health treatments, including thinning and prescribed fire, is critical to maintaining forest resilience in a warming climate (Delyser et al., 2025). Policies that align forest treatments, biomass utilization, climate goals, and greenhouse gas reduction can reduce wildfire risk while ensuring that utilization infrastructure supports, rather than distorts, restoration outcomes.

4. CONCLUSION

Taken together, these pathways point toward a future in which California's forests are once again shaped by frequent, beneficial fire rather than catastrophic megafires. Biomass utilization plays a supporting role in this transition, helping to manage the legacy of fire suppression while landscapes are brought back into balance. The long-term goal is not perpetual intervention, but the restoration of living systems capable of sustaining themselves through wise, adaptive stewardship.

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